



Choice modeling using dot-product attention mechanism

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Abstract

Modelling the cognitive process is a challenging task. Contextual conditions and the scope of the options are critical factors that influence human decisions. We propose and formulate an attention-based network to model the various choices made by humans based on the various factors that predict the possible option in each scope. To evaluate our proposed method, we conducted a user choice experiment in which a user chose an option from among limited choices. Our results showed that our model successfully extracted the hidden context on the attention layer and even outperformed the chance level in terms of prediction accuracy.

Keywords Attention network · Choice modeling · Hidden context

1 Introduction

Many researchers have been studying models of humans to recognize the world around them. There are two main types of studies: (1) models for common sense models such as semantic networks (e.g., WordNet [1] and ConceptNet [2]) and (2) models using the individual subjectivity [3] such as human preferences and impressions. An artificial system capable of deeply communicating with a person would need to understand the individual subjectivity of the person.

In this research, we focus on the choice made by a person. There are a number of methods for preference estimation; the collaborative filtering [4, 5] techniques are the typical framework for this purpose. In these methods, the estimation is based on the assumption that the preference of a person for an item is similar to the preference of similar persons for similar items. However, they are not a model of the choices made by a person, which may depend heavily on the situations encountered by that person.

In this paper, we designed a ‘choice task’ in which a person chooses one option from the given discrete options (scope) in a context, as shown in Fig. 1. Since not only the context but also the scope may affect the person’s choice [13], it is necessary to develop a mathematical model for a person’s making choices.

Recently, the dot-product attention mechanism [6–8] was proposed and used in neural network frameworks to cope with contextual information. In this mechanism, the network output was determined based on three factors: query, keys, and value. The dot-product attention mechanism performs information processing similar to a dictionary which changes its own behavior according to the training of the network. That is, the connection between the index (key) and the query and the output are tuned to fit the training data.

Since the original dot-product attention model is not suitable when the number of the available options is limited, we proposed the *masked* dot-product attention model for choice modeling where a mask is introduced to ignore unavailable options. In this model, the person’s choice is inferred as follows. The criterion fit for the situation, i.e., attention, is estimated at first. Then, the best option among the available options is chosen based on the criteria.

The experimental result suggests that the proposed model could estimate the person’s choice with a certain accuracy. Furthermore, the activation pattern of the attention in the model seemed consistent with the subjective criterion.

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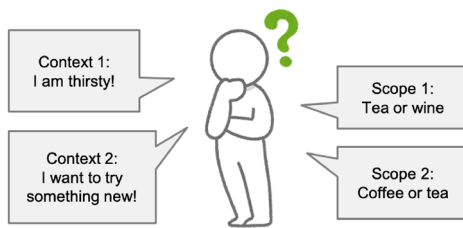


Fig. 1 Choice task in the daily environment

2 Choice tasks

The aim of this research is to model the decision-making process from the recorded behavior of the person. Instead of collecting the actual activities of the person over a long time, we built a simple graphical interface in which the person chooses an option from among the given choices. Only a limited number of options were considered in each task. This also was realistic because not all options were necessarily available. When we buy a drink from a vending machine, we need to choose one drink from the given options. Furthermore, it seems difficult for a person to simultaneously consider all possible options in the world.

The decision making would depend on the context c and the choices, that is, the possible options o to be taken, as shown in Fig. 1. Although the contexts are usually diverse in the actual case, the context in this research was defined as the situation described by a simple sentence. A finite number of contexts were prepared in advance. In this paper, c_k denotes the k -th context and $C = \{c_1, c_2, \dots, c_{|C|}\}$ denotes the set of conditions, where $|C|$ denotes the number of context. Furthermore, o_k denotes the k -th option and $O = \{o_1, o_2, \dots, o_{|O|}\}$ denotes the set of options, where $|O|$ is the number of options.

In each choice, one context was presented with several options to the person, and the person chooses one of the options. Let the ‘scope’ s be of the subset of O . Then, the task can be rephrased as choosing an option o under the condition c and the scope s .

The input to the model is defined as a combination of the context c and scope s , and they are encoded into a binary vector as shown in Fig. 2. The context of this research is an element of a finite set; therefore, it is converted into a one-hot vector i_c . The scope is a subset of O , and it is represented as the $|O|$ dimensional vector i_o . The elements in the scope are set to 1 and 0 for the other element. The input to the network is written as $i = [i_o, i_c]$ whose dimensionality is $|i| = |C| + |O|$.

To illustrate the data format, we will explain it using a simple problem setting in which $C = \{breakfast, lunch, dinner\}$ and O

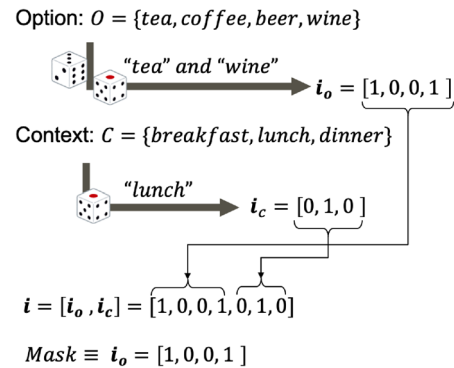


Fig. 2 Data representation

$= \{tea, coffee, beer, wine\}$. When the context is ‘lunch’ and the scope include only two options (i.e., *tea* and *wine*), then $i_c = [0, 1, 0]$ and $i_o = [1, 0, 0, 1]$. Consequently, the input becomes $i = [1, 0, 0, 1, 0, 1, 0]$.

The process of the choice was modeled as a classification model that inferred the option selected by the person. The objective of the modeling was to obtain a model for inferring the probability that the option o was chosen under the given c and s . $P(o|c, s) = 0$ for $o \notin s$. For this, we introduce ‘mask’ to the dot-product attention mechanism.

3 Masked dot-product attention model

In this section, we explain the proposed method called “Masked dot-product attention model (Fig. 3),” and its objective function used in the training.

3.1 Dot-product attention-based network

The input and output of the proposed network are the contextual input and the selected item, respectively, the selected item is represented as a one-hot vector in which the element corresponding to the selected item is 1, and the other elements are 0.

The structure of the proposed network is shown in Fig. 1. The model has several parameters, that is, the query matrix W_q , key matrix M_k , value matrix M_v , and weight parameter W_f .

First, the input vector i is multiplied with the parameter W_q , and a sigmoid activation function is applied. Then the query q , a Q dimension vector, is obtained as follows:

$$q = F_{\text{sigmoid}}(W_q \cdot i^T). \quad (1)$$

Next, the attention α (i.e., the activation of the attention layer) is calculated as follows:

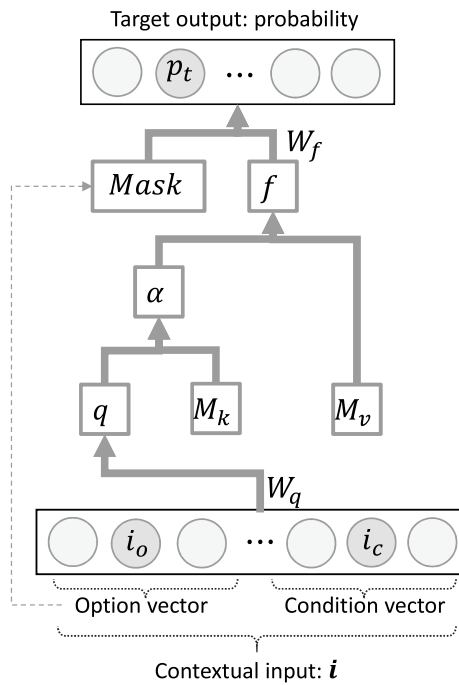


Fig. 3 Architecture of the attention network

$$\alpha = [\alpha_1, \dots, \alpha_K] = F_{\text{softmax}}(k \cdot q^\top), \quad (2)$$

where the key value $k = M_k$. The softmax function $F_{\text{softmax}}(\cdot)$ is used to normalize the attention.

Then, the embedding feature f is calculated by the multiplying the value matrix M_v with the attention α :

$$f = M_v \cdot \alpha^\top. \quad (3)$$

The embedding feature vector f is projected onto the one-hot vector space using the weight parameter W_f consists of $|O|$ elements:

$$x = W_f \cdot f^\top. \quad (4)$$

To ignore the options excluded in the given choice, we employ the mask m which indicates whether or not each option is included in the choice. The mask m is identical to the options vector $m = i_o$.

The final output o of the network is calculated as follows:

$$o = F_{\text{mssoftmax}}(x, m) = \frac{1}{Z} [m_1 e^{x_1}, \dots, m_N e^{x_N}], \quad (5)$$

where $F_{\text{mssoftmax}}(\cdot, \cdot)$ is called the masked softmax function and $Z = \sum_i m_i e^{x_i}$. Each element of the output o indicates the estimate of the probability that the corresponding option is selected.

Model parameters The model parameters are $\Theta = \{W_q, M_k, M_v, W_f\}$. When the feature f is calculated, no activation

function is used; therefore, we do not need to train both M_v and W_f . For simplicity, we updated W_q , M_k and M_v during training by keeping W_f fixed. The size of matrices W_q and M_v are $K \times Q$ and $F \times K$, respectively.

We expected that the user would choose the best option using different criteria (e.g., common sense or his/her personal values) In the proposed method, the value matrix M_v corresponds to the set of criterions, that is, the ranking of the options for the given context. The attention α helps to choose the context for the user.

3.2 Training of the model

The purpose of the training was to obtain the optimal parameters Θ^* , which would be capable of approximating the choices made by an individual:

$$P_t = [P_{t_1}, \dots, P_{t_o}] = F_{\text{softmax}}(W_f \cdot f^\top, m; \Theta^*) \quad (6)$$

Here the P_{t_k} specifies whether the option k is selected in the scope t .

A choice task is a kind of classification task; therefore, the objective function to train our model becomes:

$$\text{Loss} = L_{\text{CE}}(P_t, T) + \beta_2 L_2(\Theta) + \beta_{1/2} L_{1/2}(\alpha), \quad (7)$$

where β_2 and $\beta_{1/2}$ are the weight for each regularization.

L_{CE} represents the cross-entropy loss [9] and is calculated as follows:

$$L_{\text{CE}}(P_t, T) = - \sum_{i=1}^O t_i \log(P_{t_i}), \quad (8)$$

where P_t denotes the target, that is, the result of the choice made by the model person.

L_2 is the parameter regularization term [10] to avoid the over-fitting problem and is defined as $L_2 = |\Theta|^2$. This term is crucial when the number of data samples are not large. Since we collected the choice task results of a single user, we could only obtain only hundreds of samples.

$L_{1/2}$ is the regularization term on the attention value α and is defined as $L_{1/2} = |\alpha|^{1/2}$. It is expected that the distribution of the attention becomes sparse by this term. This might be useful since it is easy to interpret the role of the attention value when the distribution of the attention value has salient peaks, that is, it is sparse,

Derivative of the masked softmax function On the contrary to the commonly-used dot-product attention network, our proposed network has the masked softmax function. However, The derivative of the masked softmax function (Eq. (5)) is similar to the derivative of the normal softmax function and calculated as follows:

$$\frac{\partial o_j}{\partial x_k} = \begin{cases} o_j(1 - o_j) & k = j \\ -o_j o_k & k \neq j \end{cases} \quad (9)$$

Note that, the gradient for the option that does not exist in the scope is zero $(\partial o_j)/(\partial x_k) = 0$, that is, j such that $m_j = 0$ because $o_j = \frac{m_j e^{x_j}}{Z} = 0$.

Except for this part, the training can be performed in the usual way. We applied the stochastic gradient descent optimizer [11] with momentum factor. We also employed a Batch normalization technique to improved the stability of the training.

4 Experiment and evaluation

We investigated whether our proposed method could predict human choices and whether the activation pattern of the attention would change according to the context. In this section, we firstly introduce the data collection procedure. Then, we achieve the accuracy of the prediction. Furthermore, we discuss the impact of the weight parameters on the performance and the shape of the attention distribution.

4.1 User interface for a choice task

A user's choices are repeatedly collected by a web-based user interface that we developed (see Fig. 2). The task was to choose an option from among a few options (i.e., the scope). In this experiment, each option corresponded to the restraint, and the options were presented along with the condition. The user was instructed to choose a restraint from the scope by considering the condition.

The condition set contained 32 different conditions such as “Hanging out with a friend in the evening,” “Working overtime until 10 o'clock and being alone in the evening,” “With mother on weekend,” “With lover after a fireworks show,” “With co-workers after finishing system development,” and “With family on a Friday evening.” The contexts included samples that produced negative and positive feelings.

The option set contained 32 different restaurants of various types of restraint, for example, a ‘Udon-ya’ (Japanese noodle) restaurant, a Japanese style Chinese restaurant, a ‘Gyu-don’ (Japanese style beef rice) restaurant, and a convenient store. The fine-dining restaurants, such as a Chinese hotel restaurant, a Sukiyaki (Japanese hot pot) restaurant, and a French restaurant, were also included

Figure 2 shows the web-based data collection interface. The question about the context was displayed in the top of this figure and the options were listed with their photos. The user was asked to press the button on the option name to the input data.

Each time we randomly chose one condition and randomly sample four restaurant options as the scope, namely $|s| = 4$. The user was asked to input the data using the web-based interface. For training and validation, we collected 1,000 datasets from one person.

4.2 Parameter settings and training condition

In our designed task, we can derived the dimension of each layer such that $|l| = |C| + |O| = 64$ and the output $|P_t| = |O| = 32$. Other network parameters Q , K and F were set to 9, 6 and 32, respectively. Notice that $F = |P_t|$. We simply set W_f as a fixed identity matrix and trained only M_v .

4.3 Performance evaluation

Figure 4 shows the learning curves for the prediction accuracy. We used the five-fold cross-validation [12] technique to evaluate the network structure (the dimension of each layer). For each fold 800 data items are used as training samples and 800 data items are used for testing. The horizontal axes of the figures are the average training and test accuracy.

We tested the performance of the network by changing the regularization parameter $\beta_{1/2}$. The results shows that the training accuracy improves as on this parameter becomes small while the test accuracy in the 3 cases are approximately 55%. Since the four-choice chance level is (25%), the

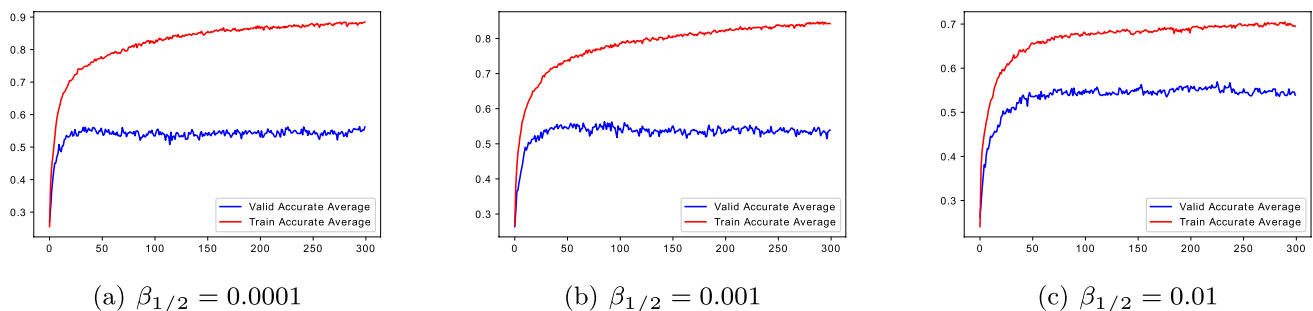


Fig. 4 Learning curves

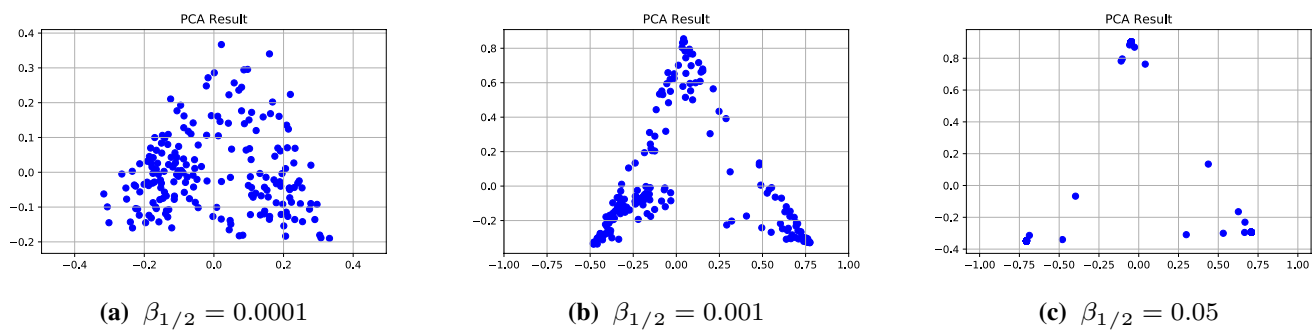


Fig. 5 Effect of regularization parameter $\beta_{1/2}$ on α

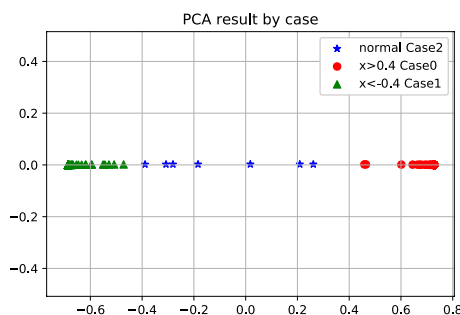


Fig. 6 Activation pattern of the attention

performance of the proposed method is significantly better than the random choice.

Figure 5 shows the impact of altering the regularization factor $\beta_{1/2}$ on the activation of attention after the learning. We trained the network with different weight values for $\beta_{1/2}$, that is, 0.0001 (left), 0.001 (middle) and 0.05 (right), and the distribution becomes more sparse when the number becomes large. Strong regularization makes the distribution sparser. On the contrary, we observed that the training prediction accuracy was dropped significantly.

Figure 6 shows the typical distribution of the attention value. Each point corresponded to the activation pattern of 200 validation datasets projected onto the two most significant components' space (possessing the largest principal component). In this case, the attention α distributes along with the first principle component (on the horizontal axis).

We divided the samples into three groups. Only a few samples (7 out of 200 data samples) are located in the middle (star). Therefore, we focus on the samples located at either end. We set two thresholds ($x < -0.4$ and $x > 0.4$). Below are the examples belonging to each case the option with the star mark (★) is the predicted option.

Low score group (< -0.4 , it triangles in Fig. 7):

1. Context: Your research paper has been accepted.
options: A*: Izakaya (Japanese bar) restaurant,



Fig. 7 The interface for data collection

B: Western restaurant, C: Restaurant providing buffet, D: Campus cafeteria

2. Context: After watching a romantic movie with your partner. options: A: Fried skewers, B: Sushi (conveyor belt), C*: Sukiyaki, D: Curry Rice
3. Context: If you are going to a matchmaking party. options: A: Horumonyaki (Japanese grilled offal) restaurant, B: Western restaurant, C: Campus cafeteria, D*: Yakitori (Japanese grilled chicken) restaurant

High score group (> 0.4 , circles in Fig. 7):

1. Context: After watching a comedy movie alone. options: A: Hot pot restaurant, B: Convenience store C*: Curry rice restaurant, D: Gyu-don restaurant
2. Context: If system development failed with your junior co-worker. options: A: Italian restaurant, B*: Japanese meals at home, C: Buffet restaurant, D: Sukiyaki restaurant
3. Context: When returning home at the usual time. options: A: Kaiseki cuisine restaurant, B*: Hamburger shop, C: Western restaurant, D: Gyu-don restaurant

In Fig. 6, the attention along the horizontal axis can be interpreted as the motivation of the user to choose a restaurant. For small value on the horizontal-axis, the user tends to choose an expensive restaurant, and for the larger value, the user tends to choose a less expensive restaurant. Although the exact meaning of this axis is not available, the attention mechanism seems to contain important latent information about the criterion of the individual user.

When the regularization parameter is large, the distribution of the attention becomes more sparse. This suggests that the diversity of the attention becomes reduced, and it becomes easy to assess the result from the viewpoint of interpretability.

4.4 Discussions

Several parameters determines the network structure; therefore, we will explain the role of these parameters. Furthermore, we will discuss the impact of the weight of the attention regularization.

Dimensionality of the matrices The dimensionality Q of the query matrix W_q , which is identical to the size of the column of the key matrix M_k . This dimensionality represents the length of the key vector, that is, the complexity of the key. The K , that is, the size of the row of the key matrix corresponds to the number of independent criteria for choosing the option. If this number is increased the network can deal with more criteria but it makes the learning difficult. We compared several patterns and found that K and Q were too small, and the learning curve may include discontinuous jumps. The design needs to carefully consider the balance between the model simplicity and the performance.

The dimensionality F represents the length of the encoding feature vector. When F was set to $|O|$, the value matrix M_v could be interpreted as a commonly used ranking table of the options, where each component corresponded to the item preference. Note that, the model has K preference tables.

The key matrix M_k and value matrix M_v can be learned by other approaches or directly connected to the input layer. In our experiment, we separated them and made them trainable parameters.

5 Conclusion and future work

In this paper, we proposed a model for making choices of individual. Using the dot-product attention mechanism, our proposed method can extract the hidden context and the criteria used by individuals for making choices. The experiment shows that our model could reach an acceptable prediction accuracy in a four-choice task. The attention layer outputs a classified pattern that matches the decision bias of the experimenter.

We will further explore this approaches to interpret the attention axis and to improve prediction accuracy. One limitation is that it is difficult to obtain much data while we found that insufficient sample size of the training data will significantly decreases the validation accuracy. It is beneficial to use data obtained from other users and to develop an hierarchical model to compensate for the lack of data.

It is also important to discuss the difference between our proposed method and existing methods such as context-aware recommendation system. One of the important difference is that the proposed method directly infers the choice make by a person. In other words, our proposed method is trained in a supervised manner whereas the commonly used recommendation systems are trained without any supervision.

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